Late Breaking Results: Evaluation of Human Action Quality with Linear Recurrent Units and Graph Attention Networks on Embedded Systems.

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ABSTRACT

Recent evolutions of recurrent neural networks (RNN) such as S4, S4D, and LRU, have shown remarkable potential for very long-range sequence modeling tasks for vision, language, and audio. They have shown a capacity to capture dependencies over tens of thousands of steps. Unlike transformers, which face significant memory consumption challenges with large context sizes, they are a promising alternative with their ability to operate effectively on embedded systems. While they have been evaluated for classification and segmentation tasks, no work in the literature has applied them in the context of human pose estimation. In this work we propose an architecture that combines such state space models (SSM) to graph attention networks (GAT) to enable their application to evaluate human action tasks on embedded systems.

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1 INTRODUCTION

Recently developed sequence modeling methodologies employing State Space Models (SSMs) have demonstrated considerable promise. Examples are S4 [4], S4D [5], and LRU [9], which have reached a state-of-the-art standard, potentially surpassing even the performance of transformers [11]. SSMs offer an advantageous combination of a swift, nonsequential training method while achieving high inference performance and low memory consumption similar to RNNs. During training, they adopt convolutions or parallel scans that can be easily implemented in embedded computing boards, such as NVIDIA Jetson, to allow even training on the edge. While transformers allow for impressive reasoning, their memory consumption scales quadratically with their context size as they need to keep all temporal data in memory to construct their attention matrix. The RNN nature of recent SSM models makes them particularly interesting for embedded systems, where RAM is generally limited. They need only to keep the current state in memory to operate and can compress the historical data without an important loss of reasoning capability even with extremely long sequences of data. Driven by these considerations, we evaluated their use for the human action evaluation in Industry 5.0 applications, such as fatigue detection of workers or recipe following in human-robot interaction. Such evaluations require even tens of thousands of frames of historical data to achieve the result, which makes transformers inapplicable for the quadratic memory requirement. Other variants of RNNs, such as LSTM [6] and GRU [2], may handle sequences of frames without quadratic memory growth. However, they are notably slow and hard to train, and still subject to vanishing and exploding gradients for large sequence lengths, hampering their reasoning ability. We propose an architecture that, differently from the state of the art solutions, combines linear recurrent unis (LRU) and graph attention networks (GAT) for evaluating prolonged human actions in real time on resource-constrained devices. We present the preliminary results conducted on a standard and widespread dataset with a Jetson Nano device.

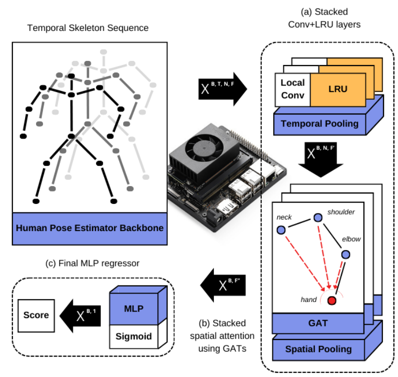

Figure 1: Architecture of the proposed model. Tensor data dimensions between components are shown in the black arrows.

2 MODEL

We adopt graph neural networks (GNN) [15] to learn the spatial relationship between human body joints and how they relate to the correct execution of actions. In particular, we adopt GAT networks [12] and combine them with the temporal reasoning in a time-then-graph layout. Figure 1 shows the proposed architecture. It starts from a sequence of skeleton data frames, X, which consists of a B, T, N, F tensor, where B is the batch dimension, T is the number of frames in the action, N is the number of skeleton joints, and F is the number of features for each joint. The skeleton data is captured by

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Table 1: Comparison with state of the art on the KIMORE dataset. Mean absolute error in the range 0-50.

Ex	Our (MAE)	Mourchid et al. [8]	Deb et al. [3]	Song et al. [10]	Zhang et al. [14]	Liao et al. [7]
1	0.928	0.641	0.799	0.977	1.757	1.141
2	1.352	0.753	0.774	1.282	3.139	1.528
3	0.846	0.210	0.369	1.105	1.737	0.845
4	0.971	0.206	0.347	0.715	1.202	0.468

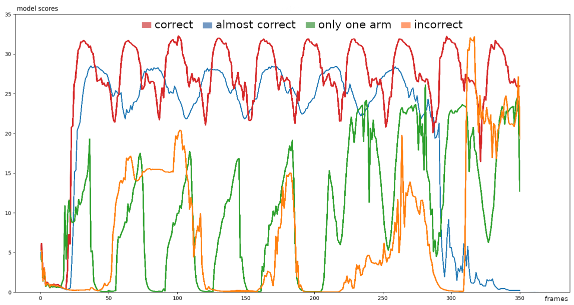

Figure 2: Real time inference of the model.

using a backbone human pose estimation software (e.g., TRTPose [13]). X is forwarded to a stack of Conv+LRU blocks, which are used for each joint independently. The convolution modules capture the local temporal information while the linear recurrent units process the entire sequence of arbitrary length using a parallel scan. We extended the Conv+LRU blocks with a standard layer normalization, dropout, and skip connection to aid the training efficiency. We then compare each joint temporal representation with the other skeleton joints to learn their spatial dependencies and importance using a stack of GATs. We compress all node's representations in a single vector with a mean pool operation, which is then passed to an MLP with a final sigmoid activation function. The output is multiplied with the maximum score possible to obtain a final, human-comprehensible score.

3 EXPERIMENTAL RESULTS

To evaluate the efficiency of the proposed architecture, we used the KIMORE dataset [1]. It contains a collection of exercises executed by multiple subjects with different types of disabilities and evaluated by experts through a score (in the range 0-50). Table 1 shows the accuracy evaluation of the proposed architecture compared to the state of the art solutions in terms of mean absolute error. The preliminary results underline that our solution achieves high accuracy without the need of sophisticated architectures and without the use of temporal attention mechanisms. They also confirm the LRU's capability to model sequence data in an actual, natural, and general use cases. We found that, compared to the other solutions that consist of around or over 1M parameters, our model relies on less than 200k parameters. This underlines potential improvements in parameter reduction, which is under investigation in our current work. The model can train for 600 epochs in under 5 minutes on the entire dataset, which is orders of magnitude faster than alternative RNN-based methods. To assess our model's ability to generalize the inference process to longer sequences, we conducted additional testing using an extended real-world dataset. This dataset includes videos depicting a subject performing various types of exercise repetitions, ranging from correct to partially correct and incorrect executions, with quality scores ranging from 0 to 35. Figure 2 illustrates the outcomes of this testing. Generally, our model effectively distinguishes precise movements from suboptimal ones. The accuracy of each action aligns with its corresponding score, although the limited labeling in the dataset results in fluctuating scores across video frames. This issue could be mitigated with more finely labeled datasets. Our model achieves a real-time regression rate of up to 100 frames per second on a Jetson Nano. For future endeavors, we intend to extend the proposed architecture to support the evaluation of various human tasks, such as fatigue detection and action segmentation, using an expanded real-world dataset.

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